

# The Complete Cubic Case and Explicit Paths in Polynomial Lemniscates

*Critical-value hubs, solved families, and the general boundary for Erdős Problem #1041*

WILL COOK\*

8 September 2026

---

## ABSTRACT

Every monic cubic whose roots lie in the open unit disc has two root occurrences joined inside  $\{|f| < 1\}$  by a broken line of length less than 2; for a squarefree cubic the endpoints are distinct. A critical point with the least critical-value modulus is the hub. A reflected-polynomial inequality puts this modulus below 1, and a Schur estimate supplies two nearby roots after normalization. The root equation controls both segments. We also give the origin as an explicit hub for monic trinomials in every degree, and develop low-critical-value and constant-factor estimates for general polynomials. The unrestricted degree problem remains open.

## 1 The complete cubic case

Erdős Problem #1041 asks whether two roots of a monic polynomial, all of whose roots lie in the open unit disc, can be joined inside  $\{|f| < 1\}$  by a curve shorter than 2. A short Euclidean segment need not stay in this set. For cubics, a critical point provides a hub at which both containment and length can be controlled.

**Theorem 1.1** (complete cubic case). *Let*

$$f(z) = \prod_{i=1}^3 (z - z_i), \quad |z_i| < 1.$$

*There are distinct indices  $i, j$  and a point  $c \in \mathbb{C}$  such that the broken line  $z_i \rightarrow c \rightarrow z_j$  lies in  $\{|f| < 1\}$  and has length strictly less than 2. It is a continuous curve of bounded variation. If  $f$  is squarefree, its endpoints are distinct values. If a root is repeated, two occurrences of that root may be joined by the constant curve.*

The proof chooses a critical point with the least critical-value modulus. A reflected-polynomial estimate makes this modulus smaller than 1. After translation and scaling, a cubic root-count estimate supplies two roots at distance at most 1 from the hub. The root equation then controls the whole of each segment, not just its endpoints.

---

\**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

*Proof.* The repeated-root case is immediate, so assume  $f$  is squarefree. Choose a critical point  $c$  for which  $|f(c)|$  is least, and write  $v = f(c)$ . The critical-value estimate below gives  $0 < |v| < 1$ . Write the other critical point, with multiplicity, as  $c + \delta$ . Integration of  $f'(c + x) = 3x(x - \delta)$  gives

$$f(c + x) = x^3 - \frac{3}{2}\delta x^2 + v.$$

Choose  $\alpha^3 = v$  and put  $B = \delta/\alpha$ . Then

$$f(c + \alpha w) = vP_B(w), \quad P_B(w) = w^3 - \frac{3}{2}Bw^2 + 1.$$

Minimality of  $|v|$  gives  $|1 - B^3/2| = |f(c + \delta)/v| \geq 1$ . The root-count estimate below supplies two distinct roots  $u, w$  of  $P_B$  in the closed unit disc. For either such root  $q$  and  $0 \leq t \leq 1$ , its root equation gives

$$P_B(tq) = 1 - t^2 - t^2(1 - t)q^3, \quad |P_B(tq)| \leq 1 - t^3 \leq 1.$$

Thus both segments from  $c$  to  $c + \alpha u$  and  $c + \alpha w$  lie in  $\{|f| < 1\}$ , since their images have modulus at most  $|v| < 1$ . Their concatenation has length

$$|\alpha|(|u| + |w|) \leq 2|v|^{1/3} < 2.$$

The two roots are distinct because  $\alpha \neq 0$ . This is the asserted piecewise affine curve; in the formal statement its length is the extended variation on  $[0, 2]$ .  $\square$

**Why a critical value is smaller than one.** The reflected-polynomial inequality, for a monic polynomial of degree  $n \geq 2$  with roots in the closed unit disc and critical points  $c_1, \dots, c_{n-1}$  listed with multiplicity, is

$$|f(c_j)| \leq \prod_{k=1}^{n-1} |1 - \bar{c}_k c_j|.$$

For a cubic, let  $c, d$  be its two critical points, with  $r = |c| \geq |d|$ . Gauss–Lucas gives  $r \leq 1$ . If  $r > 0$ , the displayed inequality implies

$$|f(c)| \leq (1 - r^2)|1 - \bar{d}c| \leq (1 - r^2)(1 + r^2) = 1 - r^4 < 1.$$

If  $r = 0$ , both critical points are zero, so  $f(z) = z^3 + f(0)$ ; Vieta and the strict root-disc hypothesis give  $|f(0)| < 1$ . Taking the smaller of the two critical-value moduli proves the needed bound. Squarefreeness makes that minimum nonzero.

**Why the normalized cubic has two small roots.** The precise algebraic estimate is

$$|1 - B^3/2| \geq 1 \implies P_B \text{ has two distinct roots in } \{|w| \leq 1\}.$$

Here the closed disc is sufficient: the strict factor  $|v| < 1$  already provides both strict conclusions in the main proof. To see the mechanism, write  $a = 3B/2$  and suppose two roots  $r, s$  of  $z^3 - az^2 + 1$  have modulus greater than 1. Vieta, with  $x = 1/r, y = 1/s$  and  $h = x + y$ , gives

$$xyh = 1, \quad ah = h^3 - 1.$$

For  $|x|, |y| < 1$ , the elementary Schur estimate is

$$|x + y - \overline{x + y}xy| < 1 - |xy|^2.$$

Indeed its left side is at most  $(1 - |y|^2)|x| + (1 - |x|^2)|y| = (|x| + |y|)(1 - |xy|)$ , and  $|x| + |y| < 1 + |xy|$ . Putting  $t = |h|^2$  and  $X = \operatorname{Re}(h^3)$  therefore gives

$$1 < t < 3, \quad X > X_0 := \frac{t^3 + 2t - 1}{2t}.$$

The remaining scalar inequality can be checked by the factorization

$$\begin{aligned} & 27\{t^3(2X^2 - t^3 - 3X + 3) - X\} - 2(t^3 - 2X + 1)^3 \\ &= (t^3 + X - 2)Q(X), \\ Q(X) &= -2t^6 + 14t^3X - 37t^3 + 16X^2 + 8X + 1. \end{aligned}$$

Both factors are positive:  $X_0 > 1$ ,

$$Q(X_0) = -\frac{(t-1)^3(t+2)(t^2+t+1)(2t^2-7t+2)}{t^2} > 0,$$

and  $Q(X) - Q(X_0) = 2(X - X_0)(7t^3 + 8X + 8X_0 + 4) > 0$ . Using  $a^3 = (h^3 - 1)^3/h^3$ , the positive expression is  $t^3(27 \operatorname{Re}(a^3) - 2|a^3|^2)$ . Consequently  $|1 - 4a^3/27| < 1$ , contradicting  $|1 - B^3/2| \geq 1$ . Finally, a repeated root of  $z^3 - az^2 + 1$  forces  $1 - 4a^3/27 = 0$ . Under the hypothesis all three roots are therefore simple, and at most one is outside the closed unit disc, as required.

The complete cubic statement is checked in [Lean](#), as are the [reflected critical-value inequality](#) and the [normalized cubic root count](#). These statements settle the cubic case. They do not supply a connector of length less than 2 for arbitrary degree. The next section gives a different explicit hub, the origin, for monic trinomials of every degree.

## 2 Monic trinomials, in every degree

**Theorem 2.1** (all-degree monic trinomials). *Let  $1 \leq m < n$  and*

$$f(z) = z^n + az^m + b,$$

*with every zero in the open unit disc. For any zero  $\zeta$ , the entire segment  $[0, \zeta]$  lies in  $\{|f| < 1\}$ . Distinct zeros  $\zeta_1, \zeta_2$  are therefore joined by the broken line  $\zeta_1 \rightarrow 0 \rightarrow \zeta_2$  of length  $\|\zeta_1\| + \|\zeta_2\| < 2$  inside the open unit lemniscate.*

*Proof.* If  $\zeta^n + a\zeta^m + b = 0$ , then

$$f(t\zeta) = b(1 - t^m) + \zeta^n(t^n - t^m).$$

Vieta gives  $|b| < 1$ . For  $0 \leq t < 1$ , the weights  $1 - t^m$  and  $t^m - t^n$  are nonnegative, so

$$|f(t\zeta)| \leq |b|(1 - t^m) + |\zeta|^n(t^m - t^n) < 1 - t^n \leq 1.$$

At  $t = 1$  the value is zero. Concatenating the two segments gives length  $|\zeta_1| + |\zeta_2| < 2$ . The displayed statement, with the broken line as a continuous curve of bounded variation and its length read as extended variation, is [Lean source](#).  $\square$

**A finite convex certificate.** For  $f(z) = \sum_{k=0}^n a_k z^k$  and  $f(\zeta) = 0$ , put  $S_j = \sum_{k=0}^j a_k \zeta^k$ . Abel summation gives

$$f(t\zeta) = \sum_{j=0}^{n-1} (t^j - t^{j+1}) S_j.$$

For  $0 \leq t < 1$ , the normalised value  $f(t\zeta)/(1 - t^n)$  lies in the convex hull of  $S_0, \dots, S_{n-1}$ . Bounds on these finitely many values therefore certify a whole segment. In the trinomial case the distinct control values are  $b$  and  $-\zeta^n$ , both strictly inside the unit disc. With more terms, the additional partial sums need their own bounds.

**Proposition 2.2** (prescribed-spoke guardrail). *There exist  $r \in (0, 1)$  for which every zero of*

$$f_r(z) = z^6 + \frac{1}{5}r^2 z^4 - \frac{1}{5}r^4 z^2 - r^6$$

*lies in the open unit disc, yet the radial spoke from the origin to the zero  $r$  leaves  $\{|f_r| < 1\}$ .*

Indeed, after  $z = rw$ , the polynomial factors as  $r^6(w^2 - 1)(w^4 + \frac{6}{5}w^2 + 1)$ , so all six roots have modulus  $r$ . However  $f_r(r/2) = -(327/320)r^6$ . Any  $r$  with  $320/327 < r^6 < 1$  proves the assertion. Only this prescribed spoke is excluded; other connectors remain available. The statement with the radius supplied is [Lean source](#).

### 3 The problem and the general regimes

**Problem 3.1** (Erdős #1041). Let  $f(z) = \prod_{i=1}^n (z - z_i)$  be monic with  $n \geq 2$  and all  $z_i$  in the open unit disc. Must two root occurrences be joined by a curve of length less than 2 inside  $\{|f| < 1\}$ ?

Repeated roots give a constant path between two listed occurrences, so the geometric question concerns squarefree polynomials. The original question is Problem 5 of Erdős–Herzog–Pirani [2, p. 139]. Pendyala [4, Thm. 1, p. 1] proves the complete degree-four case.

Theorem 4.1 treats arbitrary geometry when  $\mu \leq 13/25$ , without a root-location assumption. Its scaling corollary gives length less than  $(5/2)\mu^{1/n}$  in the open level  $(25/13)\mu$  for every squarefree monic polynomial. The independent 71/10 argument below exhibits how inverse lifts and boundary arcs spend area. The disk-family theorem gives a different sufficient condition, using a simple critical value  $v$  with  $0 < |v| < 1$  and normalised separation radius  $4/3$ . The collinear and sparse families retain the target length 2.

Two constructions organise the general estimates. Averaging inverse lifts and boundary arcs converts area into a freely selected connector. Resolving a simple saddle gives a holomorphic inverse whose Dirichlet energy controls its diameter trace. The weighted free-point theorem isolates a complementary critical-value estimate. Section 18 states the remaining selection question and the examples showing which information is insufficient.

## 4 Low critical values and a uniform path bound

The least critical value is the natural scale for the first merger of root components. A theorem at one fixed critical level therefore supplies a scale-invariant theorem for arbitrary monic polynomials. For a monic degree- $n$  polynomial put

$$K_t = \{z : |f(z)| \leq t\}, \quad \mu = \min_{f'(c)=0} |f(c)|, \quad \rho = \mu^{1/n}.$$

**Theorem 4.1** (low-critical parent regime). *Let  $f$  be squarefree and monic of degree  $n \geq 2$ , and let  $\mu$  be its least critical-value modulus. If  $\mu \leq 13/25$ , then two distinct roots are joined inside  $\{|f| < 1\}$  by a rectifiable curve of length strictly less than 2.*

**Corollary 4.2** (scale-free connection). *Every squarefree monic polynomial of degree  $n \geq 2$  has two distinct roots joined in  $\{|f| < (25/13)\mu\}$  by a curve of length less than*

$$2((25/13)\mu)^{1/n}.$$

*In every degree the length can be chosen less than  $(5/2)\mu^{1/n}$ .*

*Proof.* Put  $s = ((25/13)\mu)^{1/n}$  and  $g(z) = s^{-n}f(sz)$ . Then  $g$  is monic and its least critical-value modulus is  $13/25$ . Apply the theorem and scale the curve back by  $s$ . The rescaling step, transporting both the level and the extended variation of an arbitrary curve under  $z \mapsto h + cz$ , is [the affine transport of a contained rectifiable connector](#); the low-critical input remains ordinary. For  $n \geq 3$ ,  $(25/13)^{1/n} \leq (25/13)^{1/3} < 5/4$ . For  $n = 2$ , the segment between the roots has length  $2\sqrt{\mu}$  and lies in  $K_\mu$ .  $\square$

**The geometric step behind 13/25.** Suppose no curve of length less than 2 joins two roots below level 1. Conformal normalisation of an ancestor component gives points  $b_1, \dots, b_k$  in the hyperbolic disc with pairwise distance at least  $D$ . Write  $d_j = d_{\text{hyp}}(0, b_j)$ . The inverse-map estimates also give a lower radius  $d_* > 0$  and

$$d_j \geq d_*, \quad \sum_j \ell(d_j) \geq x := \log(t/\mu), \quad \ell(d) = -\log \tanh(d/2).$$

The open balls of radius  $D/2$  around the  $b_j$  are disjoint. Their intersections with any fixed hyperbolic circle of radius  $r$  are therefore disjoint arcs. The hyperbolic cosine law gives their angular half-widths

$$w(d, r) = \arccos \left[ \frac{\cosh d \cosh r - \cosh(D/2)}{\sinh d \sinh r} \right]_{-1}^1, \quad \sum_j w(d_j, r) \leq \pi,$$

where brackets denote truncation to  $[-1, 1]$ . Choose radii  $r_i$  and weights  $\sigma_i \geq 0$ . If

$$\ell(d) \leq U + \sum_i \sigma_i w(d, r_i) \quad (d \geq d_*), \quad U > 0,$$

then summing over the roots proves the finite dual bound

$$k \geq \frac{x - \pi \sum_i \sigma_i}{U}.$$

The summation and supremum step from the majorant and the slice bounds to the boxed inequality is [the finite dual arity floor, with the packing bound and the bound-ness of the dual values as premises](#); the hyperbolic supplier of those premises is ordinary. Angular projection alone loses the disjointness used here: balls at different radii may have overlapping shadows. Common circle slices retain it. The resulting lower bound for  $k$  feeds the length–area comparison until a short boundary connection is forced. The complete comparison proof and rational certificate are in `ErdosProblems/Erdos1041/AngularBudgetLowCriticalClosure.md`. The certified terminal time is  $635762889599/10^{12}$ , and  $(13/25) \exp(635762889599/10^{12}) < 1$ .

**An independent area proof.** The next theorem gives a weaker numerical bound through an argument that needs no circle-slice certificate. It makes the area allocation explicit.

**Theorem 4.3** (unconditional constant-factor path). *For every monic polynomial  $f$  of degree  $n \geq 2$ , two zero occurrences are joined by a possibly degenerate path of length at most*

$$\frac{71}{10} \rho$$

*inside  $K_{2\mu}$ . If  $f$  is squarefree, their locations are distinct. If every root lies in the open unit disc and  $\mu \leq 1/2$ , the construction may be chosen inside  $\{|f| < 1\}$  with length at most 5.7.*

The constant is the rationally certified specialization of a two-parameter bound. If the selected working component contains  $k \geq 2$  roots, then for every  $r \in (0, 1)$  and  $\lambda > 1$  the proof constructs a path in  $K_{\lambda\mu}$  whose length is at most

$$\sqrt{\frac{2}{k}} \left( \frac{\sqrt{2}r}{(1-r)^2} + \lambda^{1/n} \left( \sqrt{\log(\lambda/r)} + \frac{\pi}{\sqrt{\log \lambda}} \right) \right) \rho. \quad (\text{CF})$$

For  $n \geq 3$ , the choice  $\lambda = 2, r = 3/20$  makes the bracket less than  $71/10$ ; degree two has the exact root-segment bound  $2\rho$ .

*Proof.* Write  $K_t = \{|f| \leq t\}$ ,  $C_T$  for the component of  $K_T$  containing a minimising critical point, and  $A(\sigma) = \text{Area}(K_\sigma \cap C_T)$ . Degree two is the root segment of length  $2\rho$  inside  $K_\mu$ ; assume  $n \geq 3$ .

If a regular component  $C'$  of  $K_\sigma \cap C_T$  holds  $k'$  roots, then  $|dz| = \sigma d\varphi/|f'|$  on  $\partial C'$  and  $\int d\varphi = 2\pi k'$ . Cauchy–Schwarz and coarea give

$$\mathcal{H}^1(\partial C')^2 \leq 2\pi k' \sigma A'(\sigma). \quad (\text{CF1})$$

Pólya’s inequality  $A(T) \leq \pi T^{2/n}$  and the change of measure  $d\sigma/\sigma$  on  $[\mu, T]$ ,  $T = \lambda\mu$ , produce a non-critical level  $t \in [\mu, T]$  with  $tA'(t) \leq \pi T^{2/n}/\log \lambda$ . The component  $C_t \subseteq C_T$  still holds  $k \geq 2$  roots, and

$$\mathcal{H}^1(\partial C_t) \leq \pi \sqrt{2k/\log \lambda} T^{1/n}.$$

The level  $t$  is chosen from the mean of  $\sigma A'(\sigma)$  before any path is assembled. That is the only device that removes the logarithmic divergence of  $A'$  at a critical level.

For arguments avoiding the finitely many critical-value rays, each root in  $C_t$  lifts along a value-ray to  $\partial C_t$ . Split every lift at level  $r\mu$ . On a one-root lobe the inverse of  $f/\mu$  is univalent, so Koebe distortion bounds the low pieces pointwise; the precritical identity

$$\sum_i \frac{\mu^2}{|f'(z_i)|^2} \leq \rho^2 \quad (\text{CF2})$$

then aggregates them by Cauchy–Schwarz, paying  $\sqrt{2/k}$  rather than once per root. Coarea and (CF1) bound the high pieces only after averaging in the argument. Choose one argument realising that average, order the lift endpoints cyclically on  $\partial C_t$ , and hop along the shortest adjacent boundary arc. Averaging the two lifts and the  $k$  arcs over adjacent pairs gives (CF). When  $\lambda\mu \leq 1$  the same mean-value choice may be taken strictly below the top of the window, so the path lies in  $\{|f| < 1\}$ .  $\square$

The component-arity, capacity and two-root persistence-window refinements are retained with their complete hypotheses in `ErdosProblems/Erdos1041/UnconditionalConstantFactorBound.md` and `ErdosProblems/Erdos1041/MinimalHubWindowJoin.md`. Their parent conclusions under  $\mu \leq 1/2$  are already covered by Theorem 4.1; their additional content is control of a specified component or a specified descent arc. The exact distinction between a simple minimum and a unique minimum is preserved with the persistence-window argument.

## 5 Critical proximity and straight-path obstructions

The strongest source-current theorem that applies in every degree is a sharp metric selection principle at a critical point. It consumes the complete logarithmic-derivative balance rather than a degree-specific coefficient pattern.

**Theorem 5.1** (critical geometric-mean proximity). *Let  $n \geq 2$ , let  $z_1, \dots, z_n, c \in \mathbb{C}$  with  $c \neq z_k$  for every  $k$ , and suppose*

$$\sum_{k=1}^n \frac{1}{c - z_k} = 0.$$

*If  $r > 0$  is determined by*

$$r^n = \prod_{k=1}^n |c - z_k|,$$

*then there are distinct indices  $i, j$  such that*

$$|c - z_i| + |c - z_j| \leq 2r.$$

The proof chooses the two smallest distances  $d_i \leq d_j$ . The reciprocal balance gives  $d_j \leq (n-1)d_i$ , while minimality and the geometric-mean identity give  $d_i d_j^{n-1} \leq r^n$ . After normalising by  $r$ , a sharp two-variable Bernoulli inequality yields  $d_i + d_j \leq 2r$ . The complete complex theorem, including the selector and both inequalities, is checked as *exists two roots dist sum le two mul geom mean*.

This is not silently promoted to a path theorem. The same source module checks two exact barriers to the most tempting completions.

**Proposition 5.2** (two straight-path no-go families). *There is a monic quintic with all roots in the open unit disc and a non-root critical point  $c$  whose unique nearest root has a point on the straight spoke to  $c$  outside  $\{|f| < 1\}$ . There is also a monic cubic with all roots in the open unit disc such that the midpoint of every pair of distinct roots lies outside  $\{|f| < 1\}$ .*

The certificates are exact rational-algebraic computations, checked as *nearest spoke unique nearest spoke escapes* and *all straight cubic every pair midpoint escapes*, and the displayed proposition itself, with both quintic and cubic exhibited as polynomials and the nearest-root condition quantified over every zero, is [Lean source](#). The first defeats a tie-breaking escape hatch; the second defeats every straight root-pair segment. Together with Theorem 5.1, they isolate the real gap: critical-point proximity selects a short metric pair, but universal containment requires curved or topological geometry.

## 6 Solved polynomial families

Three source-current families do cross that containment gap. Their complete paper theorems are ordinary mathematics assembled from formalised load-bearing kernels; this section keeps that authority split explicit.

### 6.1 Translated cubic quotient fibres

**Theorem 6.1** (translated cubic quotient fibres). *Let  $q \geq 2$ ,  $h \in \mathbb{C}$ , and let  $P$  be a monic cubic. Put*

$$f(z) = P((z - h)^q).$$

*If all zeros of  $f$  lie in the open unit disc and  $f$  has at least two distinct zeros, then two distinct zeros of  $f$  are joined by a two-segment path of length strictly less than 2 contained in  $\{|f| < 1\}$ .*

For cubic roots  $r, s, v$ , the real charges  $\Re(rs + \overline{v})$  sum to more than  $-3$ , so one is greater than  $-1$ . For that root, AM–GM and the exact cyclotomic cancellation  $(1 - t)(1 + t + t^2) = 1 - t^3$  prove that its complete origin spoke is safe. Lean checks this fan-in as *cubic has safe root spoke*. Pulling the spoke back under  $y \mapsto y^q$  gives  $q$  safe spokes from  $h$ ; two different fibre points are joined through  $h$  with length  $2|y| < 2$ . The finite root-of-unity mean-square argument and fibre assembly are ordinary, not hidden Lean claims.

### 6.2 Primitive sparse quintics

**Theorem 6.2** (primitive sparse quintics). *Let*

$$p(z) = z^5 + az^4 + bz + c$$

*be monic with all five root occurrences in the open unit disc. Then two distinct root occurrences  $w_i, w_j$  satisfy*

$$|bw_i + c| < 1, \quad |bw_j + c| < 1,$$

*and the corresponding two radial spokes join them through the origin inside  $\{|p| < 1\}$  with total length strictly below 2. If the two occurrences coincide, the associated root-to-root path is degenerate.*



The stronger closed-disc selector has non-strict tails; if  $a \neq 0$  the selected tails are strict, while for  $a = 0$  strictness is exactly the interior-root condition. The phase-sensitive boundary theorem uses the first three Newton moments and a cubic separator. Its harmonic extension controls mixed interior/boundary configurations and shows that an unsafe interior tail contributes at most  $2/31$ . Lean checks the finite selectors *primitive boundary exists two tail energy lt one* and *primitive interior exists two tail energy lt one*; the rotation, moment identification, Abel path estimate, and two-spoke assembly are ordinary mathematics.

### 6.3 Sharp collinear roots

The complete all-degree collinear theorem appears in Section 15. It gives the sharp Chebyshev constant, not merely an existence bound, and therefore belongs to the solved-family layer rather than the frontier layer. Lean checks the constrained alternation engine *exists peak le of monic comparison* and its Chebyshev consumer *exists peak le comparison bound*; affine normalisation, root-gap selection, and transport back to the original line are ordinary.

## 7 Critical-value separation in every degree

The solved families of Section 6 are unconditional and complete, and each is confined to a shape of root configuration. This section gives a hypothesis that is not a shape at all. It constrains only the critical spectrum, it holds in every degree above an exact cutoff, and where it holds it gives the target constant 2 rather than a constant multiple of it.

**Theorem 7.1** (critical-value separation). *Let  $f$  be monic of degree  $n$ , let  $c$  be a simple critical point with  $v = f(c) \neq 0$ , and let  $S > 1$ . Suppose every other critical point  $d$  of  $f$  satisfies*

$$\left|1 - \frac{f(d)}{v}\right| \geq S.$$

*Then the square-root-resolved inverse branch through  $c$  joins two roots of  $f$  inside  $\{|f| \leq |v|\}$  by a path of length at most*

$$2|v|^{1/n}(1+S)^{1/n}\sqrt{\log \frac{S}{S-1}}.$$

*In particular the connector is shorter than  $2|v|^{1/n}$  whenever*

$$(1+S)^{2/n} \log \frac{S}{S-1} < 1. \tag{1}$$

**Corollary 7.2** (disk-family uniform radius). *A strictly stronger disk-family theorem, proved in `ErdosProblems/Erdos1041/DiskFamilyCriticalValueSeparation.md`, replaces the coefficientwise logarithm  $\log(S/(S-1))$  by the Bergman segment identity*

$$\left(\int_{-q}^q |h| \right)^2 \leq \frac{4}{\pi} \operatorname{artanh}(q^2) \|h\|_{A^2}^2$$

*and yields a uniform radius  $S = 4/3$  in every degree  $n \geq 3$ , at every real centre. The smallest degree-uniform radius certified by that coefficient at centre  $a \in [0,1]$  is  $S_\infty(a) =$*

$(A + \sqrt{A^2 - 4a(1-a)})/2$  with  $A = \coth 1$ ; the endpoints recover  $S_\infty(0) = S_\infty(1) = \coth 1$  already on file. An obstructing critical value may depend on the proposed centre. The older coefficientwise regimes  $S = 4$  ( $n \geq 3$ ),  $S = 3$  ( $n \geq 4$ ), and  $S = 2$  ( $n \geq 6$ ) remain valid and are superseded. A proposed all-degree cutoff  $S = 3/2$  for the older bound is strictly weaker than  $4/3$  and is not used.

**Corollary 7.3** (a parent-problem regime). *Let  $f$  be monic of degree  $n \geq 3$  with all roots in the open unit disc, and let  $c$  be a simple critical point with  $v = f(c)$  and  $0 < |v| < 1$ . If some real centre  $w_0 \in [0, 1]$  admits a radius  $S \geq 4/3$  such that  $|f(d)/v - w_0| \geq S$  for every other critical point  $d$ , then Erdős Problem #1041 holds for  $f$ .*

The hypothesis  $|v| < 1$  is required: normalised separation controls a connector in  $\{|f| \leq |v|\}$ , which lies in the unit lemniscate only when  $|v| < 1$ . The Fekete chain supplies  $|v| < 1$  at a minimum-modulus critical point, but an arbitrary selected critical value need not. The hypothesis is sufficient, not necessary: it does not assert that some critical value is always separated by two, and the cubic  $z^3 + (3/100)z - 3/4$  shows that no such covering statement is true.

The proof of Theorem 7.1 is the same area–capacity argument used in Section 4, run at a single saddle instead of over a merge tree. Normalise so that  $c = 0$ ,  $v = 1$ , and  $|f''(0)| \neq 0$ . The square substitution resolves the saddle, so  $f(Z(\xi)) = 1 - \xi^2$  has two local holomorphic solutions interchanged by  $\xi \mapsto -\xi$ . A finite branch point of the algebraic continuation of  $Z$  requires  $f'(d) = 0$  and  $\xi^2 = 1 - f(d)$ , and the separation hypothesis puts every such point on or outside  $|\xi| = \sqrt{S}$ ; properness of a polynomial rules out escape to infinity over a bounded value set, so the monodromy theorem continues  $Z$  holomorphically through  $|\xi| < \sqrt{S}$ . It is injective there, because  $Z(\xi_1) = Z(\xi_2)$  forces  $\xi_1^2 = \xi_2^2$ , and the two inverse sheets can meet only at an excluded branch point. Writing  $Z(\xi) = \sum_{k \geq 1} a_k \xi^k$  and fixing  $1 < R < \sqrt{S}$ , injectivity and the area formula give

$$\pi \sum_{k \geq 1} k |a_k|^2 R^{2k} = \text{Area } Z(D_R),$$

while  $f(Z(\xi)) = 1 - \xi^2$  confines the image to  $\{|f| < 1 + R^2\}$ , whose area is bounded by  $\pi(1 + R^2)^{2/n}$  for a monic  $f$ . That bound is the absolute area inequality  $\text{Area}\{|f| < T\} \leq \pi T^{2/n}$  for a polynomial with unit-modulus leading coefficient, taken from Pólya [7, printed pp. 280–282]. No relative area inequality is used here; the disposition of Dubinin’s Theorem 1 is recorded in the bibliography entry [6]. Cauchy–Schwarz then gives

$$\sum_{k \geq 1} |a_k| \leq (1 + R^2)^{1/n} \left( \sum_{k \geq 1} \frac{1}{k R^{2k}} \right)^{1/2} = (1 + R^2)^{1/n} \sqrt{\log \frac{R^2}{R^2 - 1}},$$

and termwise integration of  $|Z'|$  over  $[-1, 1]$  bounds the connector length by  $2 \sum_{k \geq 1} |a_k|$ . Letting  $R \uparrow \sqrt{S}$  proves the theorem.

**Boundary.** The analytic continuation, the univalence, the area formula, and the area–capacity inequality are ordinary mathematics; they are proved in `ErdosProblems/Erdos1041/FirstMergeCriticalValueSeparationCertificate.md`

and are not formalised. What Lean checks is the numerical half of the statement, in `ErdosProblems/Erdos1041/FirstMergeCriticalValueSeparation.lean`: the squared coefficient *first merge squared coefficient* is antitone in the degree (*first merge squared coefficient* \ antitone \ degree), a squared connector length below 4 times a coefficient below 1 forces length below 2 (*first merge* \ length \ lt \ two \ of \ squared \ bound), and the older three coefficientwise regimes hold exactly (*first merge* \ exact \ convenient \ thresholds), while the live uniform radius  $4/3$  is checked in `ErdosProblems/Erdos1041/DiskFamilyCriticalValueSeparation.lean`. The Comparator entry is `ExternalVerification1041FirstMergeCriticalValueSeparation`. Lean does not prove the analytic hypothesis  $\ell^2 \leq 4c(n, S)$  that those consumers take as input, and nothing in this section asserts the unrestricted problem or the complementary near-tie regime.

Pendyala's degree-four theorem [4, Thm. 1, p. 1] has no critical-spectrum hypothesis. The present separation criterion applies in every degree  $n \geq 3$  and supplies an explicit inverse-map length estimate.

## 8 Three exact covering obstructions

None of the following is a counterexample to Erdős #1041. Each kills a covering reading of a sufficient regime. Rational cores are checked in `ErdosProblems/Erdos1041/RevisionR2ExactCores.lean`; the ordinary arguments are in `ErdosProblems/Erdos1041/ExactObstructionsR2.md`.

**Proposition 8.1** (no separation-or covering). *Let  $f(z) = z^3 + (3/100)z - 3/4$ . Every root lies in the open unit disc, both critical points are simple, the critical values lie on distinct positive rays,  $\mu > 13/25$ , and*

$$|1 - f(c_-)/f(c_+)| < 2/375 < 2.$$

The whole proposition, including the least-modulus assertion rather than a supplied minimum, is [Lean source](#).

**Proposition 8.2** (one-root gamma bound is false). *Let  $p(z) = z^8 - (3/2)z$  and let  $C$  be the component of  $\{|p| \leq 1\}$  containing the origin. Then  $C$  contains exactly one zero and a neighbourhood of the closed disc of radius  $5/8$ , so  $\mathcal{H}^1(\partial C) > 5\pi/4$ . The constant  $\Gamma(1/4)^2/(2\sqrt{\pi})$  is at most  $(\pi/2)(1 + \sqrt{2}) < 5\pi/4$ .*

A degree-uniform one-root perimeter constant, if one exists, must be at least  $2\pi$ . For each fixed  $a > 1$ , the one-root component of  $\{|z^N - az| \leq 1\}$  contains every disc of radius  $r < 1/a$  for all sufficiently large  $N$ . A circle of radius between  $1/a$  and 1 isolates that component by Rouché's theorem. Its perimeter is therefore at least  $2\pi/a$  in the limit; now let  $a \downarrow 1$ . The family  $z^N - z$  at level 1 does not supply a one-root component. Neither general path theorem uses the rejected perimeter bound.

**Proposition 8.3** (arity does not force a capacity gap). *Let  $g(z) = z^3 - (3/400)z - 3/32$ . All roots lie in the open unit disc,  $\mu = 187/2000 \leq 1/2$ , the first-merge arity is  $k_0 = 2$ , and the ancestor component at level  $2\mu$  has normalised capacity 1.*

## 9 The Newton value equation

Away from the critical set, define the complex Newton field

$$N(z) = -\frac{f(z)}{f'(z)}.$$

Let  $z(t)$  be differentiable with  $z'(t) = N(z(t))$ , and put  $w(t) = f(z(t))$ .

**Theorem 9.1** (value equation).  $w'(t) = -w(t)$ .

The computation is one line:  $w' = f'(z) z' = f'(z) \cdot (-f(z)/f'(z)) = -f(z) = -w$ . The kernel checks it as *newton flow value has deriv at*, together with the differential form of the first integral, *newton flow scaled value has deriv at zero*:

$$\frac{d}{dt}(e^t f(z(t))) = 0, \quad \text{equivalently} \quad f(z(t)) = e^{-t} f(z(0)).$$

Observe what this says about the geometry. The value moves radially inward at exponential rate and never changes argument. The lemniscate  $\{|f| < 1\}$  is therefore forward-invariant, and the flow lines are exactly the preimages of rays from the origin.

**Corollary 9.2** (ray separation). *The values  $f$  at the endpoints of any finite Newton-flow connection lie on the same oriented ray from 0 in the value plane; the trajectory in  $z$  need not be radial. Consequently, if two critical values lie on distinct positive rays, no Newton-flow saddle connection joins the corresponding critical points.*

This is checked in consumer form: the kernel accepts the implication from the hypothesis of distinct rays to the absence of a connection. It is the statement the topology needs, and it is a genuine sharpening of the criterion used in the literature.

## 10 Arguments, not moduli

It is tempting to arrange a generic perturbation so that the critical values are pairwise distinct, or that their moduli are pairwise distinct, and to conclude that saddle connections are excluded. Neither is enough.

Two distinct critical values can lie on one ray, and two critical values with distinct moduli certainly can: the ray records the argument, and the modulus is exactly the coordinate the flow contracts. By Corollary 9.2 the invariant that excludes connections is the argument. What a perturbation must therefore achieve is pairwise distinct critical-value *arguments*, which is a condition on  $n - 1$  points modulo the circle rather than on their positions in the plane.

The cost of that condition is also checked, and it is small.

**Theorem 10.1** (ray-collision locus). *Let  $a \neq b$  be complex. Every common translation  $\beta$  for which  $a + \beta$  and  $b + \beta$  lie on the same positive ray has the form*

$$\beta = \frac{ra - b}{1 - r}, \quad r \in \mathbb{R}_{>0}, r \neq 1.$$

So each pair of critical values contributes a one-real-parameter forbidden locus in the translation plane, given in closed form (*translated same positive ray parameterization*). A finite union of such loci has empty interior, so arbitrarily small translations avoid the finitely many ray collisions. Fixed-degree lower semicontinuity in Section 18 passes a dense-generic closed-level length bound to the whole coefficient class. Quantitative forward transport of a selected curve requires additional control, but is not a premise of that existence argument.

## 11 Local coordinates for perturbative competitors

For  $f(z) = \prod_j (z - a_j)$  with  $a_j \neq 0$ , put  $r = \min_j |a_j|$  and  $p_m = \sum_j a_j^{-m}$ . On  $|z| < r$ , the exact expansion

$$\log |f(z)| = \log |f(0)| - \Re \sum_{m \geq 1} \frac{p_m}{m} z^m$$

keeps the nonlinear terms as coefficient coordinates. If  $q = |z|/r < 1$ , the tail after  $N$  has modulus at most  $nq^{N+1}/((N+1)(1-q))$ ; differentiated geometric series give the corresponding first and second derivative bounds. The complete reciprocal recurrences, zero-margin staple construction, and quadratic contact test are proved in the long record and `ErdosProblems/Erdos1041/ReciprocalNewtonExpansion.md`. Their use requires a compatible local contact; the expansion alone supplies no global root-pair selection.

## 12 A proof gap in the unrestricted argument

The March manuscript's Proposition 12 claims the following load-bearing statement. For  $u = -\log |f|$ , a connected component  $V$  of  $\{u > c\}$  carrying  $m \geq 2$  simple zeros, and the stated regularity and Morse hypotheses, it constructs, for every  $\varepsilon > 0$ , an embedded spanning tree  $G_\varepsilon \subset V$  with

$$\text{len}(G_\varepsilon) \leq \frac{1}{2\pi} \int_{2\pi}^\infty P_V(t) dt + \varepsilon. \quad (5.1)$$

The final theorem uses this estimate, so the issue below cannot be bypassed by calling the proposition auxiliary. Proposition 7 of the unattributed local revision repeats the same estimate.

The estimate itself is false. Take the Cassini polynomial  $f_a(z) = z^2 - a^2$  at  $a = 9/10$ . Its component contains the two roots at distance  $9/5$  and satisfies the relevant Morse and critical-value hypotheses. Direct level-length calculation gives the upper majorant

$$4(\sqrt{a^2 + a} - a) < \frac{41}{25}$$

for the right side of (5.1) before  $\varepsilon$ , whereas every connected set containing both roots has length at least  $9/5$ . Since  $9/5 - 41/25 = 4/25$ , choosing a smaller positive  $\varepsilon$  contradicts the assertion. Thus neither a different local saddle neighbourhood nor a perfect topological decomposition can recover the printed coefficient  $1/(2\pi)$ . The exact inequality and its abstract tree-budget contradiction are checked in `ErdosProblems.Erdos1041.CassiniTreeBudget`; the level-length majorant is an explicit analytic input, not a kernel-checked integral evaluation.

At an interior index-one critical point  $p$ , the proof invokes a Morse chart

$$u = \mu + x^2 - y^2$$

and replaces the saddle by a three-ended neighbourhood having one connected lower cross-section and two connected upper cross-sections. That local model is false as written. Because  $p \in V$  and  $V$  is open, a sufficiently small closed disc around  $p$  lies entirely in  $V$ . In that disc the full Morse chart has four sectors: two components of  $u > \mu$  and two components of  $u < \mu$ . A global component argument cannot delete one local sector from a disc already contained in  $V$ .

This independently diagnoses a proof step, but the Cassini witness above also refutes the proposition's statement. A different route might cut an adjoining regular annulus along a separatrix or regular flow arc before forming the block, retain a four-pronged saddle neighbourhood and change the assembly, or replace the local construction by the ray-cut decomposition proposed below. But no repair can retain (5.1); it must pay a positive attachment cost, select only one short pair instead of spanning every root, or use a different global metric inequality. The shorter descriptions of the same three-ended block do not repair the four-sector topology.

Corollary 9.2 supplies one independent input for a different route: distinct critical-value arguments exclude saddle-to-saddle Newton connections. It does not itself prove the compact planar decomposition, classify all orbit endpoints or provide the metric gluing estimate. Those are separate problems below.

### 13 Computational searches

Raster searches over sampled root configurations supplied candidate paths through degree ten. They supply no degree-wise upper bound. A raster path becomes a continuous certificate only after every connecting segment and its length have been enclosed. The sample counts, grid distances, and search parameters are retained in the long record. Further searches should track named obstructions: clustered critical values, shrinking attachment windows, component mergers, and distance to the unit level.

### 14 Exact symmetric benchmark: complementary chords for every binomial degree

There is a complete all-degree model family which isolates the metric geometry without pretending to select paths for arbitrary polynomials. Let  $f(z) = z^n - a$ , where  $n \geq 2$  and  $0 < |a| < 1$ . After rotation write  $a = r^n > 0$ , put  $c = \cos(\pi/n)$ , and set

$$r_* = (1 + c^n)^{-1/n}, \quad \varepsilon = (1 - r^n)^{1/n}.$$

**Theorem 14.1** (complementary binomial chords). *Two adjacent zeros of  $z^n - a$  can be joined by an explicit polygonal path inside  $\{|z^n - a| < 1\}$  of length strictly below 2. For  $r \leq r_*$  the adjacent-root chord itself works. For  $r \geq r_*$ , two radial legs and an inner adjacent crossing chord work after an arbitrarily small radial contraction. These two constructions meet at  $r = r_*$ , where the outer chord attains  $|f| = 1$  at its midpoint: equality is closed containment, not the*

open lemniscate. Open containment at and above the switch uses the inner chord after a radial contraction.

The central chord calculation is exact. On the chord between adjacent roots,

$$\max |z^n - r^n| = r^n(1 + c^n),$$

with the maximum at the midpoint; hence the first construction has precisely the threshold  $r_*$ . Above it, put  $t = \varepsilon/c$ . The path from  $r$  to  $t$  to  $t\omega$  to  $r\omega$  has its crossing chord in the closed lemniscate, and its length is

$$2r - 2\varepsilon \tan\left(\frac{\pi}{4} - \frac{\pi}{2n}\right) < 2r < 2.$$

The radius  $t$  is maximal: a larger crossing chord already escapes at its midpoint. Contracting that chord to radius  $\lambda t$  with  $0 < \lambda < 1$  makes the containment strict while keeping the two radial connections and the strict length bound. For  $n = 2$ , the adjacent-root diameter is already a strict path of length  $2r < 2$ .

The proof uses the sharp inequality

$$1 + \cos(n\theta) \leq 2 \cos^n \theta \quad (|\theta| \leq \pi/n),$$

followed by elementary chord geometry and the disk inequality that justifies the contraction. It is an authored ordinary all-degree proof with bounded numerical replay, not a Lean theorem or a priority claim. Its role is a symmetric benchmark: the exact transition says why a direct outer chord fails and why the complementary inner chord repairs it. It gives no path optimality, no theorem for arbitrary monic polynomials, and no solution of Erdős Problem 3.1.

## 15 The sharp collinear theorem: order buys an exact Chebyshev extremal

Collinearity supplies the global ordering that arbitrary complex root sets lack, and it yields a sharp all-degree solved subcase. Put

$$C_n = \frac{1}{2^{n-1} \cos^n(\pi/(2n))}.$$

**Theorem 15.1** (sharp collinear root-diameter bound). *Let  $f$  be monic of degree  $n \geq 2$ , with collinear zero occurrences of diameter  $D$ . Then some two zero occurrences are joined by their straight segment, of length at most  $D$ , on which*

$$|f| \leq C_n \left(\frac{D}{2}\right)^n.$$

*The constant is sharp: equality is attained by the affine images of the scaled Chebyshev root configuration with extreme roots at distance  $D$ .*



*Proof.* Repeated zeros give the constant path, so suppose the zeros are distinct. Translation, rotation, and scaling by  $R = D/2$  reduce to a monic real-rooted polynomial  $q$  with roots

$$-1 = y_1 < \dots < y_n = 1, \quad |f| = R^n |q| \text{ on the root line.}$$

Set  $r_n = \cos(\pi/(2n))$  and compare  $q$  with the monic polynomial

$$q_*(x) = \frac{T_n(r_n x)}{2^{n-1} r_n^n}.$$

It vanishes at  $\pm 1$  and has  $|q_*| \leq C_n$  on  $[-1, 1]$ . Choose a point  $c_i$  of maximum  $|q|$  in each adjacent gap. If every gap maximum exceeded  $C_n$ , then  $h = q - q_*$  would have the alternating signs of  $q(c_i)$  at the  $c_i$ , hence a zero between each consecutive pair, as well as zeros at  $-1$  and  $1$ . These are  $n$  distinct zeros, whereas the leading terms cancel and  $\deg h \leq n - 1$ , a contradiction. Thus one selected gap has  $|q| \leq C_n$  throughout, and scaling back proves the bound.

For sharpness take the roots of  $q_*$ , namely

$$\frac{\cos((2k-1)\pi/(2n))}{r_n}, \quad 1 \leq k \leq n.$$

Their extreme roots are  $\pm 1$ , and every adjacent gap reaches  $C_n$  at a Chebyshev extremum.  $\square$

If  $D < 2$ , the displayed level is below one and this segment lies in the open lemniscate. For  $n \geq 3$ , the same strict level holds already at  $D = 2$ ; degree two is the exact boundary case. The proof is a complete authored ordinary argument. Two load-bearing alternation and Chebyshev-comparator kernels are Lean checked, while the affine normalization and transport remain ordinary; bounded replay is regression evidence only. This proves neither that every adjacent gap is safe nor any non-collinear analogue: the real order used in the alternation count is precisely what the general problem lacks.

## 16 A Poisson identity for critical-value means

Write  $\text{FP}_m$  for the equal-weight inequality  $\sum_j (\prod_k |1 - \overline{c_k} c_j|)^{1/m} \leq m$ .

The same points can define both an analytic geometric mean and a positive Poisson mixture. Their logarithmic-derivative relation converts the desired nonlinear estimate into a diagonal Taylor-coefficient inequality.

**Theorem 16.1** (weighted free-point inequality in every degree). *Let  $c_1, \dots, c_m \in \overline{\mathbb{D}}$  and let  $w_j > 0$  satisfy  $\sum_j w_j = 1$ . Set*

$$G(z) = \prod_k |1 - \overline{c_k} z|^{w_k}.$$

*Then*

$$\sum_j w_j G(c_j)^2 \leq 1,$$

*with equality if and only if every  $c_j = 0$ . Equal weights therefore give  $\text{FP}_m$  for every  $m$ .*



*Proof.* First assume  $|c_j| < 1$  and choose analytic logarithms zero at the origin. Put

$$g(z) = \exp \sum_j w_j \log(1 - \bar{c}_j z) = 1 + \sum_{\nu \geq 1} a_\nu z^\nu, \quad P = \sum_j w_j P_{c_j},$$

where  $P_c(\zeta) = (1 - |c|^2)/|\zeta - c|^2$  and  $dm$  denotes normalised arclength. On the unit circle,

$$P = 1 - 2\Re \frac{\zeta g'}{g}.$$

Subharmonicity and Taylor orthogonality therefore give

$$\begin{aligned} \sum_j w_j G(c_j)^2 &\leq \int |g|^2 P \, dm \\ &= \int (|g|^2 - 2\Re(\zeta g' \bar{g})) \, dm \\ &= 1 - \sum_{\nu \geq 1} (2\nu - 1) |a_\nu|^2. \end{aligned}$$

All series converge on a neighbourhood of the closed disc in this case. For boundary points, replace  $c_j$  by  $rc_j$  and let  $r \uparrow 1$ . The finite left sum and every fixed Taylor coefficient converge; passing first with a finite coefficient sum proves the same inequality in the limit. Equality forces  $g = 1$ . The rational function  $\sum_j w_j \bar{c}_j / (1 - \bar{c}_j z)$  then vanishes identically. A nonzero point would give a pole with nonzero total positive weight, so all points are zero. Weighted Cauchy–Schwarz gives the equal-weight linear inequality.  $\square$

The coefficient  $a_1 = -\overline{\sum_j w_j c_j}$  gives the immediate refinement

$$\sum_j w_j G(c_j)^2 \leq 1 - \left| \sum_j w_j c_j \right|^2.$$

Thus the proof records a quantitative loss when the point centroid is nonzero. The specialised four-point and central-radius proofs are retained in the long record; the all-degree inequality above removes their cardinality restriction.

**Theorem 16.2** (critical-value mean in every degree). *Let  $f$  be monic of degree  $n \geq 2$ , with roots in a closed disc of radius  $R$ , and let  $c_1, \dots, c_{n-1}$  be its critical points with multiplicity. Then*

$$\sum_{j=1}^{n-1} |f(c_j)|^{2/(n-1)} \leq (n-1) R^{2n/(n-1)}.$$

*In particular  $(S)_n$ , the inequality  $\sum_j |f(c_j)|^{1/n} \leq (n-1)R$ , holds in every degree.*

*Proof.* Translation and scaling reduce to  $R = 1$ ; the case  $R = 0$  is immediate. Write  $q = f'/n = \prod_j (z - c_j)$  and  $q^\#(z) = \prod_j (1 - \bar{c}_j z)$ . For roots strictly inside the unit disc, on  $|\zeta| = 1$ ,

$$\Re \frac{\zeta f'(\zeta)}{f(\zeta)} = \sum_i \Re \frac{\zeta}{\zeta - z_i} > \frac{n}{2}.$$

Hence  $|f - \zeta q| < |q| = |q^\#|$  there. Gauss–Lucas makes  $q^\#$  zero-free on the closed disc, so the maximum principle gives  $|(f - \zeta q)/q^\#| < 1$  inside. At a critical point,

$$|f(c_j)| \leq \prod_k |1 - \overline{c_k} c_j|.$$

The equal-weight quadratic theorem proves the required bound. Contracting a closed-disc root configuration and taking a limit treats its boundary. Power means give  $(S)_n$ , and scaling restores  $R$ .  $\square$

The constant is attained by  $f(z) = (z - h)^n - \lambda$  with  $|\lambda| = R^n$ . These ordinary analytic proofs are recorded in `ErdosProblems/Erdos1041/FreePointQuadraticAllDegrees.md` and `ErdosProblems/Erdos1041/CentredCircleQuadrinomialConnector.md`. The circle comparison and finite Taylor-coefficient identity have checked kernels; Poisson integration and the maximum-principle assembly remain ordinary. A critical-value mean supplies no attachment or path-length estimate by itself.

## 17 The exact nonlinear lifetime currency

The merge-tree route has its own exact theorem. For each nontrivial child component  $u$ , let  $k_u$  be its number of roots and let  $r_u = \beta_u / \beta_{\text{parent}(u)} \in (0, 1]$  be its merge-scale ratio. Its contribution to the attachment age of each descendant root is

$$x_u = \frac{1}{k_u} \log \frac{1}{r_u}.$$

The edge lifetime is written

$$\lambda_k(q) = \log \frac{1 + q^{2/k}}{1 - q^{2/k}}, \quad I_k(r) = \int_r^1 \frac{dq}{q \lambda_k(q)}.$$

**Theorem 17.1** (exact attachment-age/lifetime currency). *Define*

$$\Phi(x) = \int_0^x \frac{dt}{\log(\coth t)} \quad (x \geq 0).$$

*At  $t = 0$  the integrand is understood by its continuous limiting value 0 (equivalently, the integral is improper at that endpoint). For every integer  $k \geq 1$  and  $0 < r \leq 1$ , with  $x = k^{-1} \log(1/r)$ ,*

$$I_k(r) = k\Phi(x). \tag{O1}$$

*The function  $\Phi$  is increasing and strictly convex on  $(0, \infty)$ , and*

$$\frac{\Phi(x)}{x} \rightarrow 0 \quad (x \downarrow 0). \tag{O2}$$

*Consequently, for every fixed  $k \geq 1$  and every  $c > 0$ , some  $0 < r < 1$  satisfies*

$$I_k(r) < c \frac{1}{k} \log \frac{1}{r}. \tag{O3}$$

*In particular, there is no positive universal linear conversion from attachment age to lifetime.*

*Proof.* Set  $q = e^{-kt}$ . Then  $dq/q = -k dt$ , the endpoints  $q = 1, r$  become  $t = 0, x$ , and

$$\frac{1 + e^{-2t}}{1 - e^{-2t}} = \coth t,$$

which proves (O1). Since  $\coth t$ , and hence  $\log(\coth t)$ , is strictly decreasing, the positive integrand  $1/\log(\coth t)$  is strictly increasing. This proves monotonicity and strict convexity. Moreover

$$0 \leq \frac{\Phi(x)}{x} \leq \frac{1}{\log(\coth x)} \rightarrow 0,$$

which gives (O2); taking  $r = e^{-kx}$  gives (O3).  $\square$

The identity explains why a linear charge per attachment age loses the behaviour of very young edges. Chain allocation must retain the distribution of ages and the convex function  $\Phi$ . The weighted Jensen bounds and the two-young-root selector are retained in `ErdosProblems/Erdos1041/AttachmentAgeLifetimeOrlicz.md`. They still require a geometric compatibility estimate before they produce a contained connector.

## 18 The remaining geometric selection problem

Let  $\mathcal{K}_n$  be the compact coefficient class of monic degree- $n$  polynomials with roots in the closed unit disc. Define

$$\Lambda(f) = \inf_{\gamma} \text{length}(\gamma),$$

where  $\gamma \subset \{|f| \leq 1\}$  joins two different listed root occurrences. A repeated root permits a constant curve.

**A dense generic class suffices.** The functional  $\Lambda$  is lower semicontinuous. Indeed, bounded-length competitors have uniformly bounded images and admit constant-speed parametrisations with uniformly bounded Lipschitz constants. Arzelà–Ascoli, convergence of root multisets, and uniform convergence of the polynomials on compact sets produce a limiting competitor. Its length is no larger than the lower limit of the approximating lengths, and closed sublevel containment passes to the limit. Thus a dense generic bound  $\Lambda \leq 2$  proves that bound on all of  $\mathcal{K}_n$ . For an open-disc polynomial, enclosing its roots in a disc of radius  $R < 1$  and scaling gives a path with length at most  $2R < 2$  in  $\{|f| \leq R^n\} \subset \{|f| < 1\}$ .

For  $f(z) = z^n - 1$ , the punctured unit sublevel has separate root sectors. Every path between distinct roots passes through zero, and the two radial segments attain length 2. The exact closed-class question is therefore

$$\boxed{\Lambda(f) \leq \Lambda(z^n - 1) = 2 \quad (f \in \mathcal{K}_n).}$$

The lower-semicontinuity argument is proved in `ErdosProblems/Erdos1041/GenericSufficiencyClosure.md`.

**A sufficient canonical-arc estimate.** On the generic simple-critical, ray-separated class, cutting the value plane along critical rays gives the finite inverse-sheet construction recorded in `ErdosProblems/Erdos1041/AttachmentAwareReeb.md`. Let  $L_f(c)$  be the length of the canonical two-root descent arc through a critical point  $c$ . Only critical points with  $|f(c)| \leq 1$  give admissible unit-level candidates. The sufficient metric target is

$$\min_{\substack{f'(c)=0 \\ |f(c)| \leq 1}} L_f(c) \leq 2.$$

Since  $\Lambda$  minimises over all curves, this canonical minimum is an upper bound for  $\Lambda$ . Equality between the two minimisations is not assumed. The ordinary slit-sheet construction closes its stated generic topology problem; formalising that construction remains a separate task.

**The attachment obstruction is quantitative.** For  $f(z) = z^n - r^n$ , the inverse image of the positive value ray with argument  $\theta$  approaches the saddle at zero only if its value-plane direction approaches  $\pi$ . A lift meets  $B(0, \varepsilon)$ , with  $0 < \varepsilon < r$ , precisely when

$$\cos \theta < 0, \quad |\sin \theta| < (\varepsilon/r)^n.$$

The total angular window is  $2 \arcsin((\varepsilon/r)^n)$ . An angular mean length bound therefore gives no uniform control at the shrinking set of attachable directions. A successful fixed-attachment estimate needs this angular concentration or a construction free to choose its attachment. The binomial itself is solved by radial segments.

**Which information is insufficient.** The sextic of Proposition 2.2 excludes automatic control of every prescribed root spoke once further partial sums occur. The exact Cassini example in Section 12 excludes the historical global strip-tree budget even after a local saddle model is corrected. The examples in Section 8 show that the least critical modulus and a separation-two test do not cover all polynomials, and that initial merge arity does not control capacity after subsequent mergers. The all-degree critical-value inequality controls a marginal distribution; it does not identify compatible inverse branches. The ordinary note `ErdosProblems/Erdos1041/BlaschkePowerCriticalSpectra.md` records Blaschke-power polynomials  $F_N = A^N - D^N$  that realise the sharp critical-value moment numerator  $4s$  inside the polynomial class, including a high-critical nonseparated regime with exact rational certificates. The same family has ordinary short connectors of length  $O(1/N)$  and an exact degree-dependent degeneration of  $\Lambda$ . These statements are not parent proofs, are not Lean-checked as analytic theorems, and do not replace the trinomial flagship.

The remaining parent regime has  $\mu > 13/25$  and lies outside the solved families and the applicable separation conditions. Further progress requires a contained pair selected with its geometric cost, a quantitative attachment estimate, or a different competitor for  $\Lambda$ . The failed scalar implications and their exact witnesses are preserved in the long record.

## 19 Proof boundaries

The trinomial radial inequalities and length budget, the two-root proximity selector, the exact spoke counterexamples, and the specified finite kernels are checked in Lean at their linked source revisions. Their geometric assemblies are identified separately in the corresponding proofs.

The 13/25 theorem is ordinary analysis with an exact rational certificate. The 71/10 theorem, the analytic inverse-map construction, the all-degree Poisson argument, and the generic slit-sheet construction are ordinary proofs. The finite coefficient identity and the polar circle comparison have checked kernels; neither substitutes for Poisson integration or the maximum principle. The complete declaration-to-claim map is retained in the long record and the formal-source guide below.

Fixed-degree lower semicontinuity reduces the closed class to a dense generic metric bound. The admissible canonical-arc length estimate remains open. Raster searches supply candidate paths for finite samples; their meaning is separate from the exact rational certificates used in proved inequalities. No independent human review or resolution of the general problem is claimed.

### Sources and adjacent results

The original root-to-root problem is [2, p. 139, Problem 5]. Pendyala’s degree-four theorem [4, Thm. 1, p. 1] is directly comparable. His marked-point-to-boundary problem [5] has different endpoints and does not imply the root-pair conclusion. The exact Cassini obstruction in Section 12 concerns the metric assertion of [3, Prop. 12]. Downloaded-source digests, precise locators, and withdrawal history are retained in the long record. No uniqueness or priority claim is inferred from this comparison.

## A Guide to the formal sources

The public `ErdosProblems.Erdos1041.NewtonFlowRaySeparation` module contains the checked source for this note. The search of §13 is `scripts/search_counterexample.py` in the source package. The declaration table below is pinned to the shared formal-source commit used throughout this problem-note series.

- [newton flow vector](#)
- [derivative mul newton flow vector](#)
- [newton flow value has deriv at](#)
- [newton flow scaled value has deriv at zero](#)
- [same positive ray](#)
- [same positive ray refl](#)
- [same positive ray symm](#)
- [same positive ray trans](#)
- [translated same positive ray parameterization](#)

- real affine line
- real affine line eq affine span
- is closed real affine line
- dense compl real affine line
- exists small avoiding finite real affine lines
- same positive ray imp mem real affine line
- exists small translation separating arguments
- norm lt one of near root
- constant perturbation root near original
- constant perturbation roots in unit disk
- same positive ray of real exp decay
- no newton connection of not same positive ray

## References

- [1] T. F. Bloom, *Erdős Problems*, problem 1041. <https://www.erdosproblems.com/1041>
- [2] P. Erdős, F. Herzog, and G. Piranian, *Metric properties of polynomials*, J. Analyse Math. **6** (1958), 125–148. <https://doi.org/10.1007/BF02790232>
- [3] shtuka, *A Short Path Joining Two Zeros Inside a Polynomial Lemniscate*, manuscript posted 24 March 2026, 48 pp. <https://shtuka123.github.io/1041/main.pdf>
- [4] V. S. Pendyala, *A Degree-Four Lemniscate Path Theorem*, arXiv:2606.24875v1 (2026). <https://arxiv.org/abs/2606.24875>, doi:10.48550/arXiv.2606.24875.
- [5] V. S. Pendyala, *Shortest paths in polynomial lemniscate sublevel sets and a problem of Erdős*, arXiv:2606.19178v1 (2026), Theorem 1.2, p. 3, and introductory discussion, p. 2. <https://arxiv.org/abs/2606.19178>, doi:10.48550/arXiv.2606.19178.
- [6] V. N. Dubinin, *Some inequalities for polynomials and rational functions associated with a lemniscate*, Zap. Nauchn. Sem. POMI **404** (2012), 83–99; English translation, J. Math. Sci. **193** (2013), no. 1, 45–54, doi:10.1007/s10958-013-1432-4. This neighbouring covering theorem is discussed in the long record; the area input used here is the absolute Pólya inequality [7].
- [7] G. Pólya, *Beitrag zur Verallgemeinerung des Verzerrungssatzes auf mehrfach zusammenhängende Gebiete*, Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse (1928), printed pp. 228–232 and 280–282. <https://archive.org/details/sitzungsbericht1928preu>.